

Performance Report

BNO085 - STM32F767ZI - ROS2 Sensor Pipeline

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1. Purpose

This report validates the fixed-rate performance of the embedded-to-ROS2 heading pipeline and summarizes the Kalman-filter parameter selection used in the final implementation. **The main validation targets are a stable 10 Hz embedded output rate, a matching ROS2 publish rate, and measurable reduction of stationary heading noise.**

2. System Under Test

BNO085 -> SPI -> STM32F767ZI -> Kalman X/Y -> Heading -> NMEA HDM -> UART -> ROS2 /imu/data

Runtime configuration: BNO085 calibrated magnetic-field report, 10 Hz runtime output, USART3 at 115200 baud, ROS2 Humble LifecycleNode.

3. Fixed-Rate Performance Validation

Method. The embedded application was configured for a 10 Hz runtime heading/NMEA output. Timing statistics were collected after the system entered normal operation; startup/calibration transition samples were excluded so they would not distort the runtime measurement.

Metric	Result
Target frequency	10 Hz
Target period	100 ms
Average period	99.974 ms
Minimum period	99 ms
Maximum period	101 ms
Peak-to-peak jitter	2 ms
Measured frequency	10.003 Hz

Result. The average period remained within 0.026 ms of the 100 ms target. The measured interval stayed between 99 ms and 101 ms, corresponding to 2 ms peak-to-peak jitter. The embedded implementation therefore satisfies the intended fixed 10 Hz update rate.

4. ROS2 Publish-Rate Validation

Method. The final ROS2 output rate was measured with ros2 topic Hz /imu/data. The embedded NMEA stream was the only input source during this test.

Source	Measured Rate
Embedded NMEA output	10.003 Hz
ROS2 /imu/data	~10.000 Hz
Target	10 Hz

Result. The ROS2 publish rate closely matched the embedded output rate; no meaningful rate loss was observed across the UART-to-ROS2 pipeline.

5. Kalman Filter Parameter Selection and Validation

Filter structure. Two independent scalar Kalman filters are applied to the calibrated magnetic X and Y components before heading is calculated with $\text{atan2}(Y, X)$. Filtering X/Y rather than heading directly avoids the 359 deg / 0 deg wrap-around problem.

Parameter	Final Value	Selection Method
Q	0.25 uT^2/s	Experimental tuning
Rx	0.433 uT^2	Welford stationary variance
Ry	0.393 uT^2	Welford stationary variance
P0	1.0 uT^2	Initial uncertainty
K	Dynamic	Calculated at every update

R - measurement noise. The sensor was kept stationary after calibration and calibrated magnetic X/Y samples were characterized with the Welford online variance algorithm. The measured variances were used directly as Rx and Ry, avoiding an arbitrary noise setting.

Q - process noise. Q was tuned experimentally to balance stationary noise suppression and responsiveness. The final value was reduced from 1.0 to 0.25 uT^2/s after comparing recorded stationary results.

P0 and K. P0 = 1.0 uT^2 represents only the initial state uncertainty; covariance is updated during operation. Kalman gain is not a fixed tuning constant and is recalculated each update as $K = P_{\text{pred}} / (P_{\text{pred}} + R)$.

Measurement workflow. STM32CubeMonitor was connected through ST-LINK/SWD to observe runtime variables without modifying the UART data path. Raw and filtered heading signals were logged, exported as CSV files, and evaluated offline for the final stationary-noise comparison.

Stationary heading result

Metric	Raw Heading	Kalman-Filtered Heading
Mean	50.002 deg	50.007 deg
Standard deviation	2.829 deg	1.021 deg
Peak-to-peak	17.029 deg	5.818 deg

Noise reduction: standard deviation decreased by approximately 63.9%, while peak-to-peak variation decreased by approximately 65.8%. The mean remained essentially unchanged, indicating improved stability without a meaningful steady-state bias.

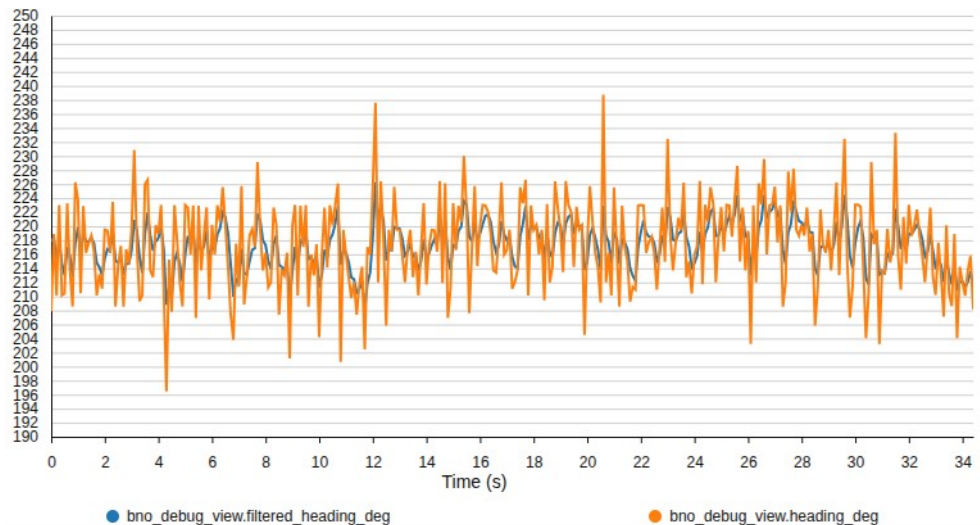


Figure 1. Representative stationary capture from STM32CubeMonitor: raw heading (orange) and Kalman-filtered heading (blue). Statistical values above were computed from the exported CSV validation dataset.